

Qiankun Zhong, PhD

Postdoc Researcher
Max Planck Institute for Human Development

Cultural Evolution, Organizational Communication,
Machine/AI Behaviors, Computational Social Science

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Blog: [self-made modeler](#)

Education

Ph.D. in Communication
Oct 2023

University of California, Davis

Advisor: Martin Hilbert

Dissertation: *Institutions and Culture in the current Communication Landscape*

M.A. in Emerging Media Studies
Sep 2015 - June 2017

Boston University

B.A in Journalism
Sep 2010 – Sept 2014

Shandong University

Additional training:
**Graduate Workshop of
Computational Social Science**
June 2022

Santa Fe Institute
Computational Modeling

Additional training:
**Summer Institute in Survey
Research Techniques**
July 2019

University of Michigan
Multi-level Analysis of Survey Data

Employment

Post-doctoral Researcher, Nov 2023 –
Center for Humans and Machines, Max Planck Institute for Human Development
Supervisor: Iyad Rahwan

Selected Publications and Preprints

Zhong, Q., Eisenmann, T., & Rahwan, I. (In Press for Philosophical Transaction of Royal Society Proceedings A). Cumulative Cultural Evolution and Multi-Level Selection of AI Behaviors. <https://arxiv.org/abs/2602.03541>

Gupta, P*., **Zhong, Q.***, Yakura, H.*, Eisenmann, T., & Rahwan, I. (2026). The Role of Social Learning and Collective Norm Formation in Fostering Cooperation in LLM Multi-Agent Systems. *The 25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2026)*. <https://www.ifaamas.org/Proceedings/aamas2026/pdfs/CZDC3237.pdf>

Zhong, Q., Jacoby, N., Tchernichovski, O., & Frey, S. (Preprint). Institutional Preferences in the Laboratory. [arXiv preprint arXiv:2502.06748](https://arxiv.org/abs/2502.06748).

Zhong, Q., Hilbert, M., & Frey, S. (2023). Breaking the Structural Reinforcement: An Agent-based Model on Cultural Consumption and Social Relations. *Social Science Computer Review*. <https://doi.org/10.1177/08944393211056501>

Zhong, Q., Frey, S. (2022). Institutional similarity drives cultural similarity among online communities. *Scientific Reports* 12, 18982. <https://doi.org/10.1038/s41598-022-23223-8>

Zhong, Q., Frey, S., & Hilbert, M. (2022). Quantifying the Selective, Stochastic, and Complementary Drivers of Institutional Evolution in Online Communities. *Entropy*, 24(9), 1185. <https://doi.org/10.3390/e24091185>

Full Publications

Lorenz, J., Engljaehring, H., Oswald, Y., **Zhong, Q.** (In Press for Historical Social Research) Emergence, Dynamics, and Implications and Wealth Inequality Misperceptions, and Wealth Taxation in an Agent-based Model. doi 10.5281/zenodo.21645474

Smaldino, P. E., Russell, A., Zefferman, M., Donath, J., Foster, J., Guilbeault, D., Hilbert, M., Hobson, E. A., Lerman, K., Miton, H., Moser, C., Lasser, J., Schmer-Galunder, S., Shapiro, J., **Zhong, Q.**, & Patt, D. (2025). Information Architectures: A Framework for Understanding Socio-Technical Systems. *NPJ Complexity*. doi: 10.31219/osf.io/c7vrw

Chakraborti, M*, **Zhong, Q.***, Bulat, B., Sen, A., & Frey, S. (Preprint). Deconstructing written rules and hierarchy in peer produced software communities. *arXiv preprint arXiv:2206.07992*.

Rao, A., Chang, R. C., **Zhong, Q.**, Wojcieszak, M., & Lerman, K. (2025) Tracking a Year of Polarized Twitter Discourse on Abortion. *International AAAI Conference on Web and Social Media*. doi: <https://doi.org/10.1609/icwsm.v19i1.35893>

Chang, R. C., Rao, A., **Zhong, Q.**, Wojcieszak, M., & Lerman, K. (2023) #RoeOverturned: Twitter Dataset on the Abortion Rights Controversy. *International AAAI Conference on Web and Social Media*. doi: <https://doi.org/10.1609/icwsm.v17i1.22207>

Cummings, J, Cahill, T. J., Wertz, B., **Zhong, Q.** (2022). Psychological Predictors of Virtual Reality Technology Adoption and Usage. *Virtual Reality*, 1-23. <https://doi.org/10.1007/s10055-022-00736-1>

Frey, S., **Zhong, Q.**, Bulat, B., Weisman, W., Liu, C., Fujimoto, F., & Wang, H. (2022). Governing online goods: An Institutional Comparison of Minecraft, Reddit, and World of Warcraft. In *Proceedings of The 25th ACM Conference On Computer-Supported Cooperative Work And Social Computing (CSCW)*. <https://doi.org/10.1145/3555191>

Cahill, T. J., Wertz, B., **Zhong, Q.**, Parlato, A., Donegan, J., Forman, R., ... & Cunningham, T. N. (2019). The Search for Consumers of Web-Based Raw DNA Interpretation Services: Using Social Media to Target Hard-to-Reach Populations. *Journal of medical Internet research*, 21(7), e12980. <https://doi.org/10.2196/12980>

Groshek, J., Katz, J. E., Andersen, B., Cutino, C., & **Zhong, Q.** (2018). Media use and antimicrobial resistance misinformation and misuse: Survey evidence of information channels and fatalism in augmenting a global health threat. *Cogent Medicine*, 5(1), 1460898. <https://doi.org/10.1080/2331205X.2018.1460898>

Wang, C., Cahill, T. J., Parlato, A., Wertz, B., **Zhong, Q.**, Cunningham, T. N., & Cummings, J. J. (2017). Consumer use and response to online third-party raw DNA interpretation services. *Molecular Genetics & Genomic Medicine*. <https://doi.org/10.1002/mgg3.340>
 *equal contribution

Research Appointment	
Research Assistant PI: Seth Frey 2020 – 2023	University of California, Davis Department of Communication Project: Designing Smart Environments to Augment Collective Learning & Creativity (NSF Award 2421385) • Implement online large-scale asynchronous economic games • Analyze experiment data using Bayesian Structural Models Project: Jumpstarting Successful Open-Source Software Projects with Evidence-Based Rules and Structures (NSF Award 2020751) • Conducted text analysis using multiple natural language processing models • Conducted data analysis using multi-level regression • Data visualization the results using R and Python • Wrote multiple preprints and published papers on CSCW and Scientific Reports
Research Assistance PI: Jim Cummings 2015-2016	Boston University School of Communication, Emerging Media Studies Project Perception of Presence in a Virtual Environment • Conducted meta-analysis on objective measurement of presence • Designed survey for VR usage and adoption • Prepared clean dataset for statistical modeling

Skills	
Programming	Python, R, Java, SQL, NetLogo, HTML, Javascript
Research	Agent-based modeling, multi-agent system, evolutionary game theory, social network analysis, econometrics, large-scale online experiment design
Language	Chinese, English, Italian

Fellowships, Awards, and Honors	
Dean’s Graduate Summer Fellowship, 2023	
School of Letters and Science, University of California, Davis	
Dean’s Graduate Summer Fellowship, 2022	
School of Letters and Science, University of California, Davis	
Student Travel Award, 2021	

Sunbelt Network Conference

Graduate Student Travel Award, 2020

Graduate Student Association, University of California, Davis

Top paper award, 2020

Zhong, Q. & Frey, S. (May, 2020) "Institutional similarities drive cultural similarity among online communities", 70th International Communication Association Annual Meeting (ICA), Online.
Computational Method, International Communication Association

Provost's Fellowship in the Arts, Humanities, and Social Science, 2018

School of letters and Science, University of California, Davis

Top poster award, 2018

Wertz, B. & Zhong, Q. (May, 2018) "Learning, Doing, Flow, and Fun: Understanding Relationships Between Difficulty, Enjoyment and Skill Development in Games" 68th International Communication Association Annual Meeting (ICA), Prague, Czech.
Game Studies, International Communication Association

Courses Taught

Introduction to Mass Communication

2021- 2022, Spring Semester

Teaching Assistant (lecture section) and Full Lecturer (lab section)

Responsibility: Lecture on basic empirical methods used in mass communication research; Grade quizzes, exams, and essays.

Data Visualization

2020- 2021, Fall Semester

Teaching Assistant (lecture section) and Full Lecturer (lab section)

Responsibility: Lecture on statistics and data visualization for social science students across communication, political science, psychology, and education; Grade exams and essays.

Introduction to Computational Social Science

2018-2021

Head Teaching Assistant and course design assistant

Responsibility: Lecture on computational social science theory and methods; Re-design course content to keep up-to-date with new changes in social science; Design comprehensive assessment to evaluate student performances.

Digital Technology and Social Change

2018-2021

Head Teaching Assistant

Responsibility: Coordinate with teaching assistants and professor to host office hours, grade homework and exams, support students with final paper writing.

Invited Talks and Guest Lectures

"Evolutionary Approach to Institutions and Culture", Human Behavior, Ecology, and Culture, Max Planck Institute for Evolutionary Anthropology, May, 2026. Leipzig, Germany.

“Group Selection as a Safeguard against AI Substitute”, Institute for Cognitive Science and Technology, National Research Council. Mar 2026. Rome, Italy.

“Institutional Evolution in the Laboratory”, Department of Psychology, Cornell University, Sep 2025, Ithaca, USA.

“Cumulative Cultural Evolution and Group Selection of AI Behaviors”, Dynamics of Social Behavior Research Group, IT:U Austria, July 2025, Linz, Austria.

“Cultural Evolution of Online Communities” USC-SJTU Institute of Cultural and Creative Industry. Shanghai Jiao Tong University. Dec, 2024. Shanghai, China.

Lecture “Comparing Python and Java in writing Agent-based Modeling” Schank ABM Lab, Nov, 2024 UC Davis, California.

“Group Selection of AI Behaviors” Computational Auditory Perception Group, Max Planck Institute for Empirical Aesthetics. June, 2024. Frankfurt, Germany.

Lecture “Agent-based Models in Communication Research”, Department of Communication, Clemson University, Nov, 2023. South Carolina, USA.

“Polarization of Beliefs and Identity Signals: comparing origins and maintenance” Information Architecture workshop, Santa Fe Institute. May, 2023. Santa Fe, USA.

Service

Panel chair on ESCoP-CoPM Symposium 2026
Organize and Moderate “Decision-making, learning, and adaptation in human–AI hybrid collectives” Panel

Conference Committee on Machine+Behavior Conference 2026
Review abstracts and organize conference program

Organizing EHBEA [AI]volution Track 2026
Review and organize sessions for [AI]volution Theme
Organize and Moderate “AI and Evolutionary Science” Panel

Professional Affiliations

- International Communication Association (ICA)
- Cultural Evolution Society

Review

- Nature Human Behavior
- Nature Communication
- Collective Intelligence
- Communication and Change
- Proceedings of the Royal Society B: Biological Sciences
- Digital Journalism
- ACM CSCW
- Cognitive Science
- Adaptive Behavior
- Computational Communication Research